with the same number of channels for horizontal and vertical coordinates of the input X along. This can be written as

$$z^h = \text{GAP}^h(X) \tag{109}$$
$$z^w = \text{GAP}^w(X) \tag{110}$$
$$f = \delta(\text{BN}(\text{Conv}_1^{1\times1}([z^h; z^w]))) \tag{111}$$
$$f^h, f^w = \text{Split}(f) \tag{112}$$
$$s^h = \sigma(\text{Conv}_h^{1\times1}(f^h)) \tag{113}$$
$$s^w = \sigma(\text{Conv}_w^{1\times1}(f^w)) \tag{114}$$
$$Y = Xs^hs^w \tag{115}$$

where GAP^h and GAP^w denote pooling functions for vertical and horizontal coordinates, and $s^h \in \mathbb{R}^{C\times1\times W}$ and $s^w \in \mathbb{R}^{C\times H\times1}$ represent corresponding attention weights.

Using coordinate attention, the network can accurately obtain the position of a targeted object. This approach has a larger receptive field than BAM and CBAM. Like an SE block, it also models cross-channel relationships, effectively enhancing the expressive power of the learned features. Due to its lightweight design and flexibility, it can be easily used in classical building blocks of mobile networks.

3.6.8 DANet

In the field of scene segmentation, encoder-decoder structures cannot make use of the global relationships between objects, whereas RNN-based structures heavily rely on the output of the long-term memorization. To address the above problems, Fu et al. [10] proposed a novel framework, the *dual attention network* (DANet), for natural scene image segmentation. Unlike CBAM and BAM, it adopts a self-attention mechanism instead of simply stacking convolutions to compute the spatial attention map, which enables the network to capture global information directly.

DANet uses in parallel a position attention module and a channel attention module to capture feature dependencies in spatial and channel domains. Given the input feature map X, convolution layers are applied first in the position attention module to obtain new feature maps. Then the position attention module selectively aggregates the features at each position using a weighted sum of features at all positions, where the weights are determined by feature similarity between corresponding pairs of positions. The channel attention module has a similar form except for dimensional reduction to model cross-channel relations. Finally the outputs from the two branches are fused to obtain final feature representations. For simplicity, we reshape the feature map X to $C \times (H \times W)$ whereupon the overall process can be written as

$$Q, \quad K, \quad V = W_qX, \quad W_kX, \quad W_vX \tag{116}$$
$$Y^{\text{pos}} = X + V\text{Softmax}(Q^TK) \tag{117}$$
$$Y^{\text{chn}} = X + \text{Softmax}(XX^T)X \tag{118}$$
$$Y = Y^{\text{pos}} + Y^{\text{chn}} \tag{119}$$

where $W_q, W_k, W_v \in \mathbb{R}^{C\times C}$ are used to generate new feature maps.

The position attention module enables DANet to capture long-range contextual information and adaptively integrate similar features at any scale from a global viewpoint, while the channel attention module is responsible for enhancing useful channels as well as suppressing noise. Taking spatial and channel relationships into consideration explicitly improves the feature representation for scene segmentation. However, it is computationally costly, especially for large input feature maps.

3.6.9 RGA

Unlike coordinate attention and DANet, which emphasise capturing long-range context, in *relation-aware global attention* (RGA) [101], Zhang et al. stress the importance of global structural information provided by pairwise relations, and uses it to produce attention maps.

RGA comes in two forms, *spatial RGA* (RGA-S) and *channel RGA* (RGA-C). RGA-S first reshapes the input feature map X to $C \times (H \times W)$ and the pairwise relation matrix $R \in \mathbb{R}^{(H\times W)\times(H\times W)}$ is computed using

$$Q = \delta(W^QX) \tag{120}$$
$$K = \delta(W^KX) \tag{121}$$
$$R = Q^TK \tag{122}$$

The relation vector r_i at position i is defined by stacking pairwise relations at all positions:

$$r_i = [R(i,:); R(:,i)] \tag{123}$$

and the spatial relation-aware feature y_i can be written as

$$Y_i = [g^c_{\text{avg}}(\delta(W^\varphi x_i)); \delta(W^\phi r_i)] \tag{124}$$

where g^c_{avg} denotes global average pooling in the channel domain. Finally, the spatial attention score at position i is given by

$$a_i = \sigma(W_2\delta(W_1y_i)) \tag{125}$$

RGA-C has the same form as RGA-S, except for taking the input feature map as a set of $H \times W$-dimensional features.

RGA uses global relations to generate the attention score for each feature node, so provides valuable structural information and significantly enhances the representational power. RGA-S and RGA-C are flexible enough to be used in any CNN network; Zhang et al. propose using them jointly in sequence to better capture both spatial and cross-channel relationships.

3.6.10 Self-Calibrated Convolutions

Motivated by the success of group convolution, Liu et at [120] presented *self-calibrated convolution* as a means to enlarge the receptive field at each spatial location.

Self-calibrated convolution is used together with a standard convolution. It first divides the input feature X into X_1 and X_2 in the channel domain. The self-calibrated convolution first uses average pooling to reduce the input size and enlarge the receptive field:

$$T_1 = \text{AvgPool}_r(X_1) \tag{126}$$

where r is the filter size and stride. Then a convolution is used to model the channel relationship and a bilinear interpolation operator Up is used to upsample the feature map:

$$X_1' = \text{Up}(\text{Conv}_2(T_1)) \tag{127}$$

Next, element-wise multiplication finishes the self-calibrated process:

$$Y_1' = \text{Conv}_3(X_1)\sigma(X_1 + X_1') \tag{128}$$

Finally, the output feature map of is formed:

$$Y_1 = \text{Conv}_4(Y_1') \tag{129}$$
$$Y_2 = \text{Conv}_1(X_2) \tag{130}$$
$$Y = [Y_1; Y_2] \tag{131}$$

Such self-calibrated convolution can enlarge the receptive field of a network and improve its adaptability. It achieves excellent results in image classification and certain downstream tasks such as instance segmentation, object detection and keypoint detection.

3.6.11 SPNet

Spatial pooling usually operates on a small region which limits its capability to capture long-range dependencies and focus on distant regions. To overcome this, Hou et al. [124] proposed *strip pooling*, a novel pooling method capable of encoding long-range context in either horizontal or vertical spatial domains.

Strip pooling has two branches for horizontal and vertical strip pooling. The horizontal strip pooling part first pools the input feature $F \in \mathcal{R}^{C\times H\times W}$ in the horizontal direction:

$$y^1 = \text{GAP}^w(X) \tag{132}$$

Then a 1D convolution with kernel size 3 is applied in y to capture the relationship between different rows and channels. This is repeated W times to make the output y_v consistent with the input shape:

$$y_h = \text{Expand}(\text{Conv1D}(y^1)) \tag{133}$$

Vertical strip pooling is performed in a similar way. Finally, the outputs of the two branches are fused using element-wise summation to produce the attention map:

$$s = \sigma(\text{Conv}^{1\times 1}(y_v + y_h)) \tag{134}$$
$$Y = sX \tag{135}$$

The strip pooling module (SPM) is further developed in the mixed pooling module (MPM). Both consider spatial and channel relationships to overcome the locality of convolutional neural networks. SPNet achieves state-of-the-art results for several complex semantic segmentation benchmarks.

3.6.12 SCA-CNN

As CNN features are naturally spatial, channel-wise and multi-layer, Chen et al. [50] proposed a novel *spatial and channel-wise attention-based convolutional neural network* (SCA-CNN). It was designed for the task of image captioning, and uses an encoder-decoder framework where a CNN first encodes an input image into a vector and then an LSTM decodes the vector into a sequence of words. Given an input feature map X and the previous time step LSTM hidden state $h_{t-1} \in \mathbb{R}^d$, a spatial attention mechanism pays more attention to the semantically useful regions, guided by LSTM hidden state h_{t-1}. The spatial attention model is:

$$a(h_{t-1}, X) = \tanh(\text{Conv}_1^{1\times 1}(X) \oplus W_1 h_{t-1}) \tag{136}$$
$$\Phi_s(h_{t-1}, X) = \text{Softmax}(\text{Conv}_2^{1\times 1}(a(h_{t-1}, X))) \tag{137}$$

where $\oplus$ represents addition of a matrix and a vector. Similarly, channel-wise attention aggregates global information first, and then computes a channel-wise attention weight vector with the hidden state h_{t-1}:

$$b(h_{t-1}, X) = \tanh((W_2\text{GAP}(X) + b_2) \oplus W_1 h_{t-1}) \tag{138}$$
$$\Phi_c(h_{t-1}, X) = \text{Softmax}(W_3(b(h_{t-1}, X)) + b_3) \tag{139}$$

Overall, the SCA mechanism can be written in one of two ways. If channel-wise attention is applied before spatial attention, we have

$$Y = f(X, \Phi_s(h_{t-1}, X\Phi_c(h_{t-1}, X)), \Phi_c(h_{t-1}, X)) \tag{140}$$

and if spatial attention comes first:

$$Y = f(X, \Phi_s(h_{t-1}, X), \Phi_c(h_{t-1}, X\Phi_s(h_{t-1}, X))) \tag{141}$$

where $f(\cdot)$ denotes the modulate function which takes the feature map X and attention maps as input and then outputs the modulated feature map Y.

Unlike previous attention mechanisms which consider each image region equally and use global spatial information to tell the network where to focus, SCA-Net leverages the semantic vector to produce the spatial attention map as well as the channel-wise attention weight vector. Being more than a powerful attention model, SCA-CNN also provides a better understanding of where and what the model should focus on during sentence generation.

3.6.13 GALA

Most attention mechanisms learn where to focus using only weak supervisory signals from class labels, which inspired Linsley et al. [122] to investigate how explicit human supervision can affect the performance and interpretability of attention models. As a proof of concept, Linsley et al. proposed the *global-and-local attention* (GALA) module, which extends an SE block with a spatial attention mechanism.

Given the input feature map X, GALA uses an attention mask that combines global and local attention to tell the network where and on what to focus. As in SE blocks, global attention aggregates global information by global average pooling and then produces a channel-wise attention weight vector using a multilayer perceptron. In local attention, two consecutive 1×1 convolutions are conducted on the input to produce a positional weight map. The outputs of the local and global pathways are combined by addition and multiplication. Formally, GALA can be represented as:

$$s_g = W_2\delta(W_1\text{GAP}(x)) \tag{142}$$
$$s_l = \text{Conv}_2^{1\times 1}(\delta(\text{Conv}_1^{1\times 1}(X))) \tag{143}$$
$$s_g^* = \text{Expand}(s_g) \tag{144}$$
$$s_l^* = \text{Expand}(s_l) \tag{145}$$
$$s = \tanh(a(s_g^* + s_l^*) + m \cdot (s_g^* s_l^*)) \tag{146}$$
$$Y = sX \tag{147}$$

where $a, m \in \mathbb{R}^C$ are learnable parameters representing channel-wise weight vectors.

Supervised by human-provided feature importance maps, GALA has significantly improved representational power and can be combined with any CNN backbone.

3.7 Spatial & Temporal Attention

Spatial & temporal attention combines the advantages of spatial attention and temporal attention as it adaptively selects both important regions and key frames. Some works [16], [130] compute temporal attention and spatial attention separately, while others [131] produce joint spatiotemporal attention maps. Further works focusing on capturing pairwise relations [177]. Representative spatial & temporal attention attentions and specific process $g(x)$ and $f(g(x), x)$ described as Eq. 1 are summarised in Tab. 8. We next discuss specific spatial & temporal attention mechanisms according to the order in Fig. 4.

3.7.1 STA-LSTM

In human action recognition, each type of action generally only depends on a few specific kinematic joints [130]. Furthermore, over time, multiple actions may be performed. Motivated by these observations, Song et al. [130] proposed a joint spatial and temporal attention network based on LSTM [147], to adaptively find discriminative features and keyframes. Its main attention-related components are a spatial attention sub-network, to select important regions, and a temporal attention sub-network, to select key frames. The spatial attention sub-network can be written as:

$$s_t = U_s \tanh(W_{xs} X_t + W_{hs} h^s_{t-1} + b_{si}) + b_{so} \tag{148}$$

$$\alpha_t = \text{Softmax}(s_t) \tag{149}$$

$$Y_t = \alpha_t X_t \tag{150}$$

where X_t is the input feature at time t, U_s, W_{hs}, b_{si}, and b_{so} are learnable parameters, and h^s_{t-1} is the hidden state at step $t-1$. Note that use of the hidden state h means the attention process takes temporal relationships into consideration.

The temporal attention sub-network is similar to the spatial branch and produces its attention map using:

$$\beta_t = \delta(W_{xp} X_t + W_{hp} h^p_{t-1} + b_p). \tag{151}$$

It adopts a ReLU function instead of a normalization function for ease of optimization. It also uses a regularized objective function to improve convergence.

Overall, this paper presents a joint spatiotemporal attention method to focus on important joints and keyframes, with excellent results on the action recognition task.

3.7.2 RSTAN

To capture spatiotemporal contexts in video frames, Du et al. [16] introduced *spatiotemporal attention* to adaptively identify key features in a global way.

The spatiotemporal attention mechanism in RSTAN consists of a spatial attention module and a temporal attention module applied serially. Given an input feature map $X \in \mathbb{R}^{D \times T \times H \times W}$ and the previous hidden state h_{t-1} of an RNN model, spatiotemporal attention aims to produce a spatiotemporal feature representation for action recognition. First, the given feature map X is reshaped to $\mathbb{R}^{D \times T \times (H \times W)}$, and we define $X(n, k)$ as the feature vector for the k-th location of the n-th frame. At time t, the spatial attention mechanism aims to produce a global feature l_n for each frame, which can be written as

$$\alpha_t(n, k) = w_\alpha \tanh(W_h h_{t-1} + W_x X(n, k) + b_\alpha) \tag{152}$$

$$\alpha^*_t(n, k) = e^{\gamma_\alpha \alpha_t(n,k)} / \sum_{j=1}^{W \times H} e^{\gamma_\alpha \alpha_t(n,k)} \tag{153}$$

$$l_n = \sum_{k=1}^{H \times W} \alpha^*_t(n, k) X(n, k) \tag{154}$$

where γ_α is introduced to control the sharpness of the location-score map. After obtaining frame-wise features $\{l_1, \ldots, l_T\}$, RSTAN uses a temporal attention mechanism to estimate the importance of each frame feature

$$\beta_t(n) = w_\beta \tanh(W'_h h_{t-1} + W_l l(n) + b_\beta) \tag{155}$$

$$\beta^*_t(n) = e^{\gamma_\beta \beta_t(n)} / \sum_{j=1}^{T} e^{\gamma_\beta \beta_t(n)} \tag{156}$$

$$\phi_t = \sum_{n=1}^{T} \beta^*_t(n) l(n) \tag{157}$$

The spatiotemporal attention mechanism used in RSTAN identifies those regions in both spatial and temporal domains which are strongly related to the prediction in the current step of the RNN. This efficiently enhances the representation power of any 2D CNN.

3.7.3 STA

Previous attention-based methods for video-based person re-identification only assigned an attention weight to each frame and failed to capture joint spatial and temporal relationships. To address this issue, Fu et al. [131] propose a novel *spatiotemporal attention* (STA) approach, which assigns attention scores for each spatial region in different frames without any extra parameters.

Given the feature maps of an input video $\{X_n | X_n \in \mathbb{R}^{C \times H \times W}\}_{n=1}^N$, STA first generates frame-wise attention maps by using the l_2 norm on the squares sum in the channel domain:

$$g_n(h, w) = \frac{|| \sum_{c=1}^{C} X_n(c, h, w)^2 ||_2}{\sum_{h=1}^{H} \sum_{w=1}^{W} || \sum_{c=1}^{C} X_n(c, h, w)^2 ||_2} \tag{158}$$

Then both the feature maps and attention maps are divided into K local regions horizontally, each of which represents one part of the person. The spatial attention score for region k is obtained using

$$s_{n,k} = \sum_{(i,j) \in \text{Region}_k} ||g_n(i, j)||_1 \tag{159}$$

To capture the relationships between regions in different frames, STA applies l_1 normalization to the attention scores in the temporal domain, using

$$S(n, k) = \frac{s_{n,k}}{\sum_{n=1}^{N} ||s_{n,k}||_1} \tag{160}$$